

Correlation Illusion Cards Student Handout

TOK Cognitive Science Activity Bank • Mathematics • 25-35 min

Category	Mathematics	Time	25-35 min
Difficulty	Easy	ID	correlation-illusion-cards

Big idea: Patterns can feel meaningful before they are mathematically justified.

Student-Facing Prompt

Inspect Correlation Illusion Cards Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Correlation Illusion Cards Stimulus A	Student-facing stimulus 1: Cards pair 'used flashcards / did not use flashcards' with 'high quiz score / lower quiz score'. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Correlation Illusion Cards Stimulus B	Student-facing stimulus 2: A few vivid cards show flashcards plus high scores, while the full set is mixed. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Use this for comparison, a second round, or a partner challenge.
Correlation Illusion Cards Reveal / Twist	Student-facing stimulus 3: A few vivid cards show flashcards plus high scores, while the full set is mixed. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Teacher reveal: connect the changed response to the big idea — Patterns can feel meaningful before they are mathematically justified.

Actual Lesson Questions

#	Question for students
1	After inspecting Correlation Illusion Cards Stimulus A, what is your first knowledge claim?
2	Which exact detail from Correlation Illusion Cards Stimulus A did you use as evidence?
3	What did you infer beyond the evidence?
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?
5	What missing information would most improve the knowledge claim?
6	When does a perceived pattern become evidence?

Ready-to-Use Examples

- Cards pair 'used flashcards / did not use flashcards' with 'high quiz score / lower quiz score'.
- A few vivid cards show flashcards plus high scores, while the full set is mixed.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Mathematics.

Activity Steps

1. Show a small set of memorable paired examples and ask students to guess the relationship.
2. Give groups the full card set to sort into a four-cell table.
3. Students count all cells, not just confirming cases.
4. Groups decide whether the perceived relationship survives counting.
5. Debrief why humans notice some pairings more than others.

Observation and Evidence Record

Use this grid to keep observation, inference, evidence, and confidence separate.

What I noticed	What I inferred	Evidence	Confidence

TOK Debrief

- Knowledge question: When does a perceived pattern become reliable mathematical evidence?
- Why are confirming cases often more memorable than disconfirming cases?
- How should knowers protect themselves from pattern hunger?

Knowledge Questions

- When does a perceived pattern become evidence?
- How do salience and sample size affect knowledge?
- Can counting change what seems obvious?

Transfer Example

For example, run this activity with cards with random pairings that appear patterned after selective attention. Have students commit to an initial claim, then reveal the change, hidden rule, alternative context, or comparison that makes the knowledge problem visible.

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...